Homework: Static & Kinetic Friction Practice

Name: ______________________________ **Date:** __________________

1. A 14 kg box rests on a level surface. The coefficient of static friction is $\mu_s = 0.40$. A student pushes horizontally on the box.

1. Draw a Free Body Diagram (FBD).
2. Find the weight F_g.
3. Find the normal force F_N.
4. Find the maximum static friction $F_{f_{s,\max}}$.
5. If the applied push is 30 N, does the box move? Explain.
6. What is the friction force acting on the box?
7. What is $\sum F_x$?
8. What is $\sum F_y$?
9. What is the acceleration in the x-direction?

2. A 10 kg box is pushed horizontally with increasing force. The coefficients of friction are $\mu_s = 0.55$ and $\mu_k = 0.25$.

1. Draw a Free Body Diagram (FBD).

2. Find the weight F_g.

3. Find the normal force F_N.

4. Find the minimum applied force required to start motion. (Hint: minimum applied force = maximum static force)

5. If the applied force is 60 N. Is the box moving? Why or why not?

6. What is the friction force acting on the box?

7. What is $\sum F_x$?

8. What is $\sum F_y$?

9. What is the acceleration in the x-direction?

3. A 18 kg box is pushed to the right with a constant horizontal force of 85 N. The coefficients of friction are $\mu_s = 0.45$ and $\mu_k = 0.30$.

1. Draw a Free Body Diagram (FBD).

2. Find the weight F_g.

3. Find the normal force F_N.

4. Determine whether the box starts moving. Show work by comparing the applied force to $f_{s,\max}$.

5. What is the friction force acting on the box?

6. What is $\sum F_x$?

7. What is $\sum F_y$?

8. If the box moves, find the acceleration of the box.

4. A box rests on a rough horizontal surface. A student pushes on the box with a horizontal force that slowly increases from 0 N.

1. Describe how the **static friction force** changes as the applied force increases but the box remains at rest.

2. What happens when the applied force becomes equal to the **maximum static friction**?

3. Explain why the box can remain at rest even though a horizontal force is applied. (Hint: Think of Newton's Laws)

4. Once the box begins to slide, how does the friction force change? Why is this force usually smaller than the maximum static friction?